Web Services

Multi-Tier Architecture

(MP1 - GL)

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Table of Contents

1 Multi-Layer Architecture

2 Transaction Processing

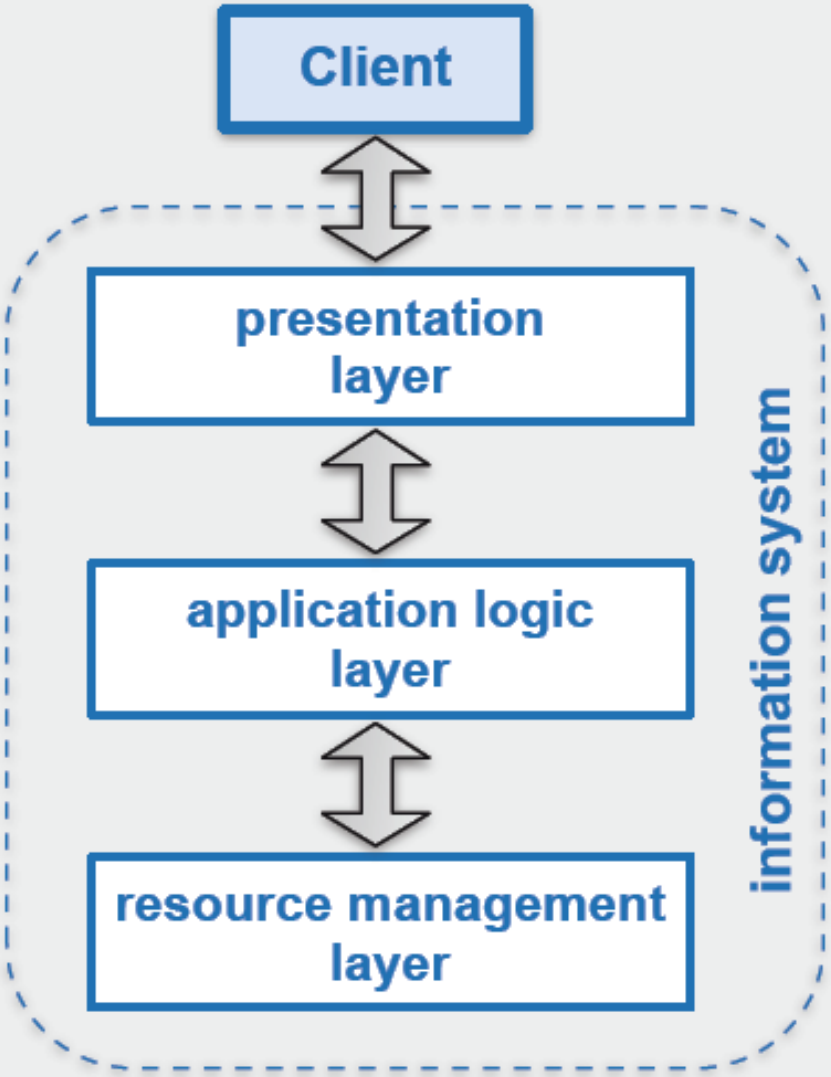
3 Multi-Tier Architecture

Multi-Layer Architecture

Multilayered Software Architecture

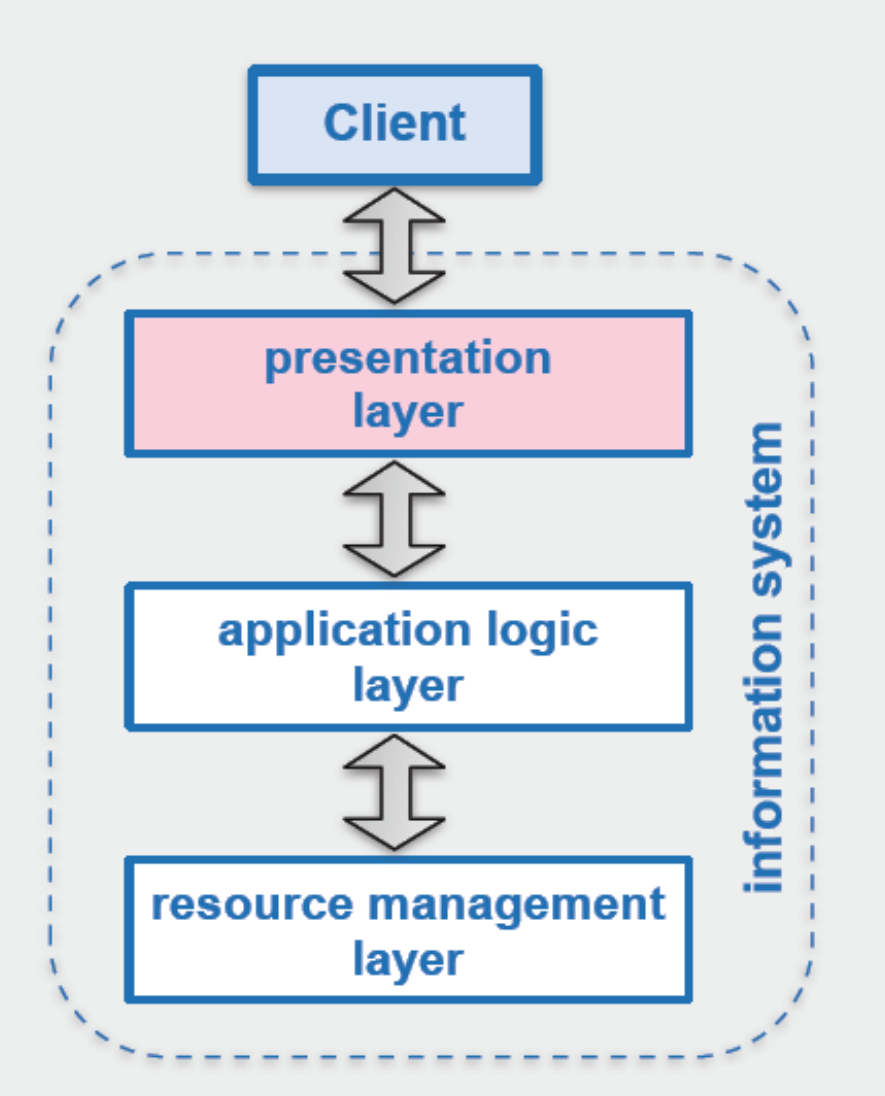
- A *multilayered* software architecture is a software architecture that uses many layers for allocating the different responsibilities of a software product.
- "**Layer**" represents the orientation of the different physical or conceptual elements that make up an entire software solution.
- "**Tier**" represents the physical layout of the various mechanisms in a system's infrastructure.

3 Layers Of an Information System



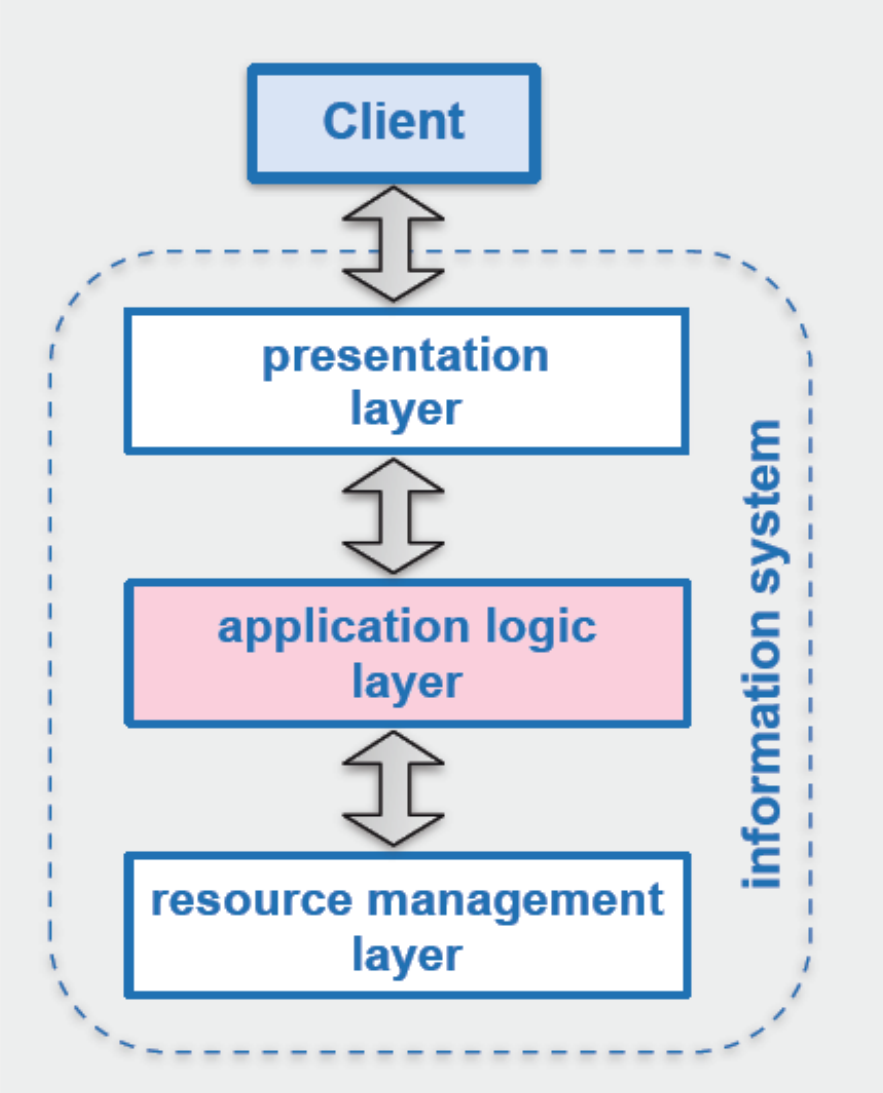
Presentation Layer

- Controls how the information system presents information to external entities and accepts it from them.
- External entities are users (UI) or other information systems (API)



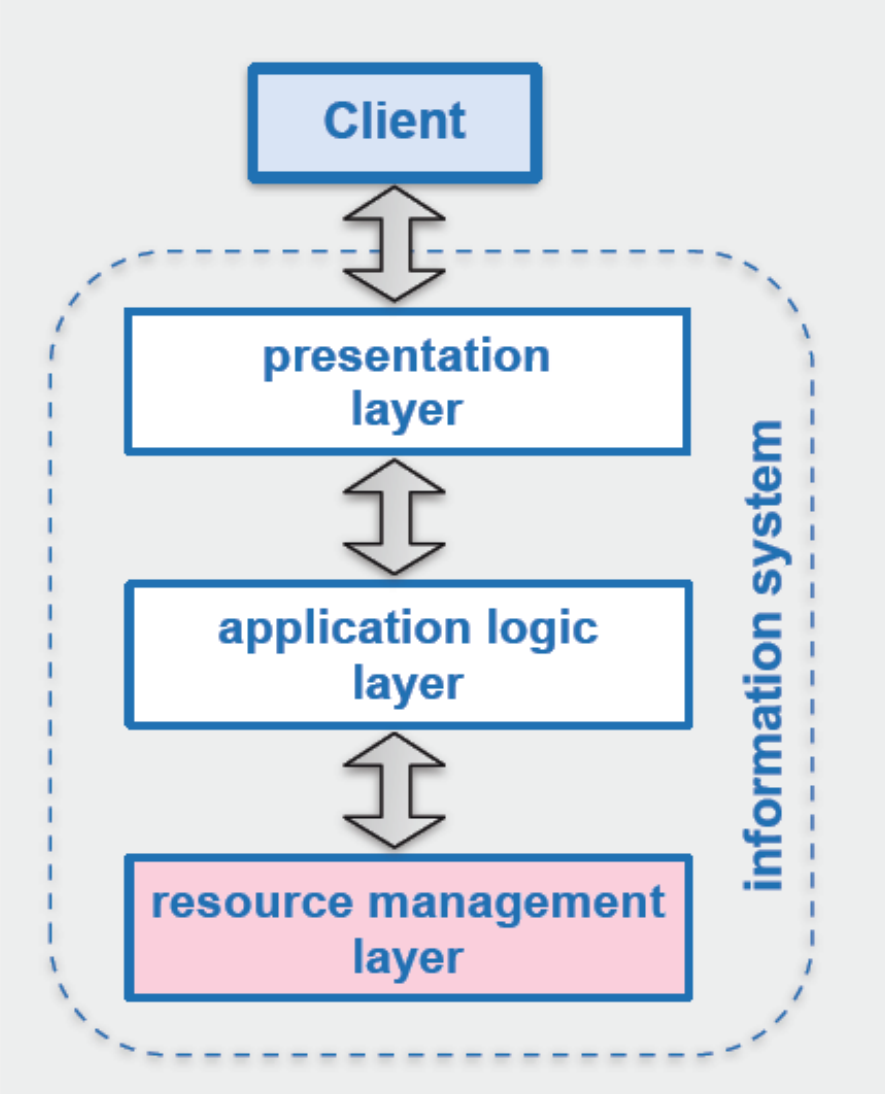
Application Logic Layer

- The program
- Business process
- Business logic
- Business rules



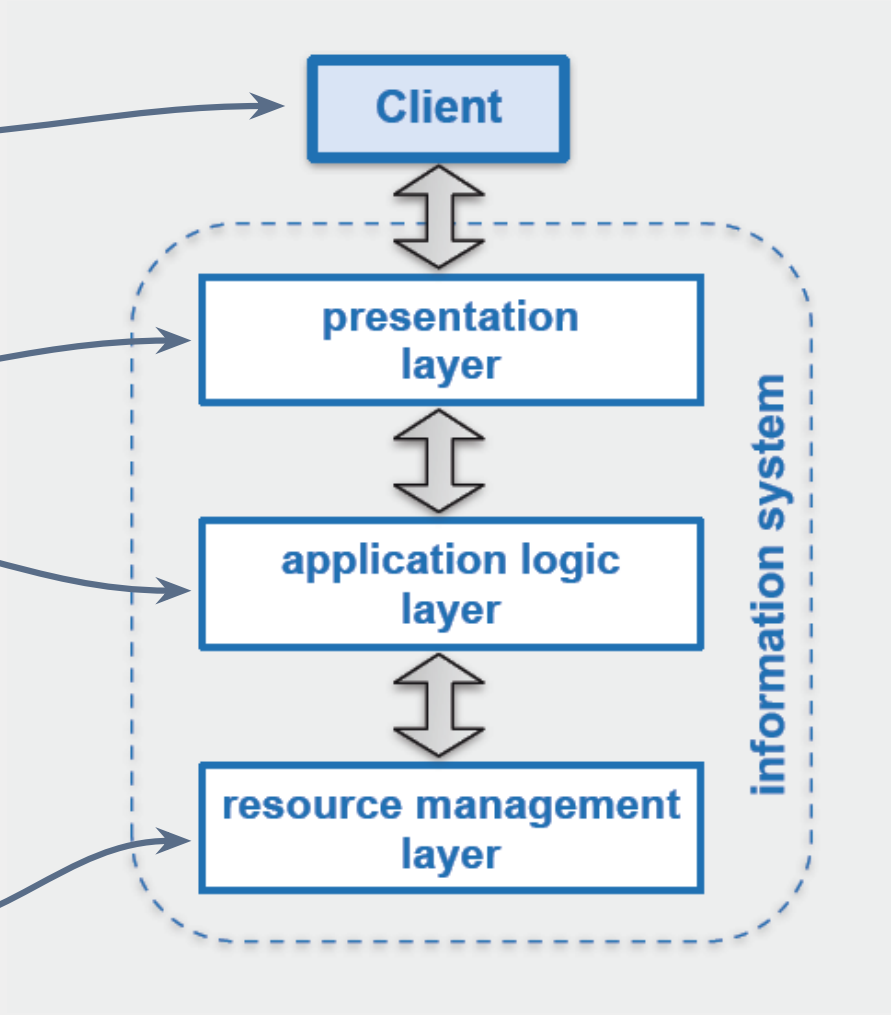
Resource Management Layer

- Implements the resource manager
- Takes care of ACID properties
- We will discuss them in coming slides



Top-Down Design

1. define access channels and client platforms
2. define presentation formats and protocols for the selected clients and protocols
3. define the functionality necessary to deliver the contents and formats needed at the presentation layer
4. define the data sources and data organization needed to implement the application logic



Bottom-Up Design

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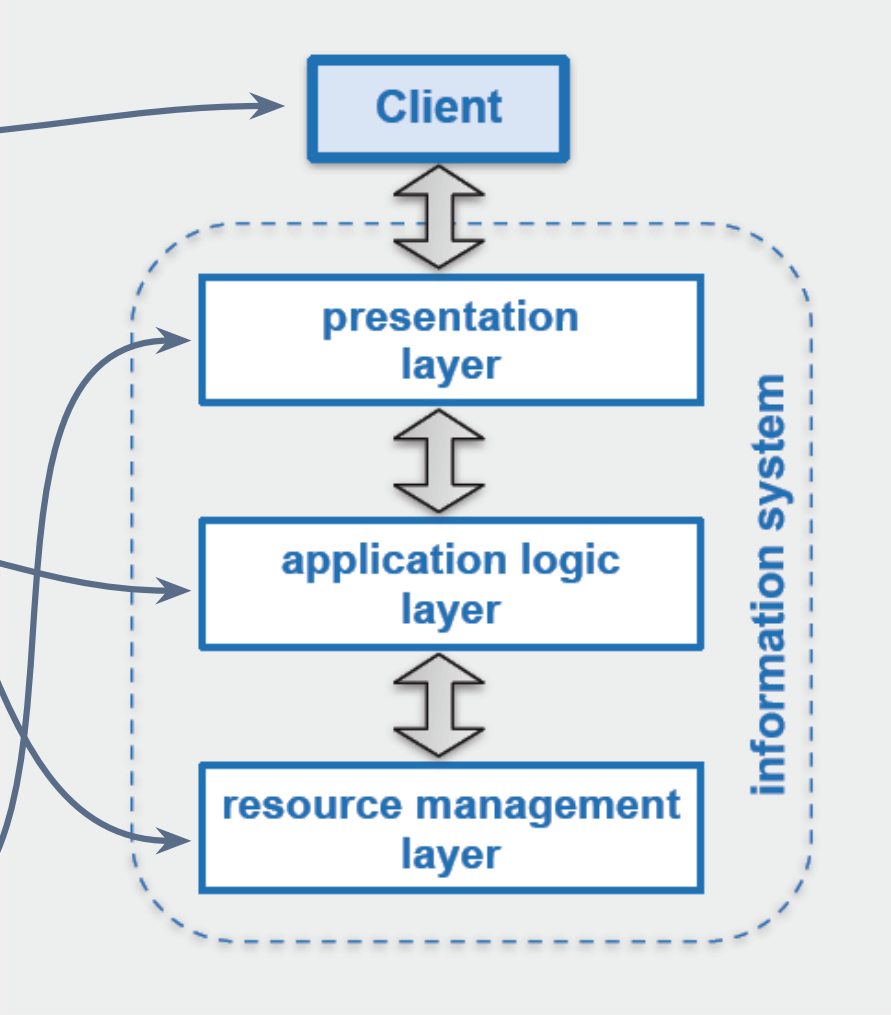

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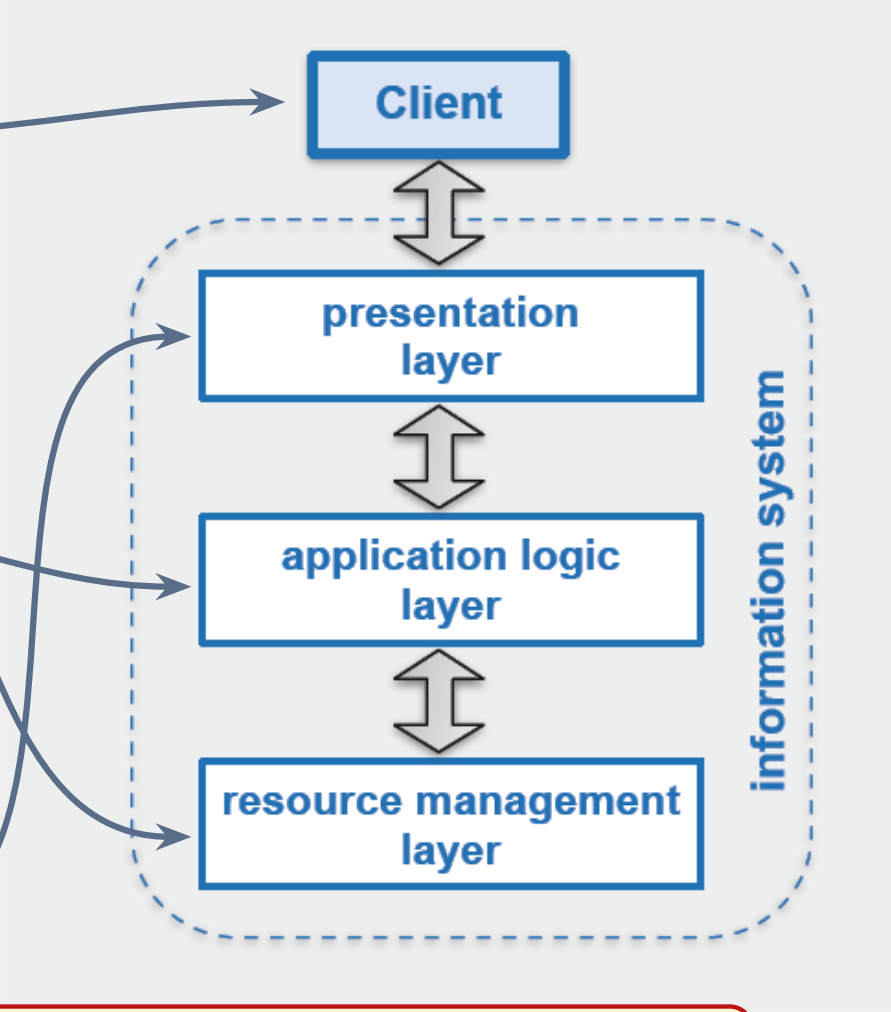


Note: Used for Legacy Systems

Transaction Processing

Transaction Processing

- Business Transaction
  - Interaction in real world
  - Usually between enterprise & person or between enterprises
- Information processing that is divided into individual, indivisible operations, called transactions performs function on (shared) database
- OnLine Transaction Processing (OLTP)
  - Runs a collection of transaction programs online

The ACID Properties

- A set of properties that guarantee that transactions are processed reliably
  - Atomicity
  - Consistency
  - Isolation
  - Durability

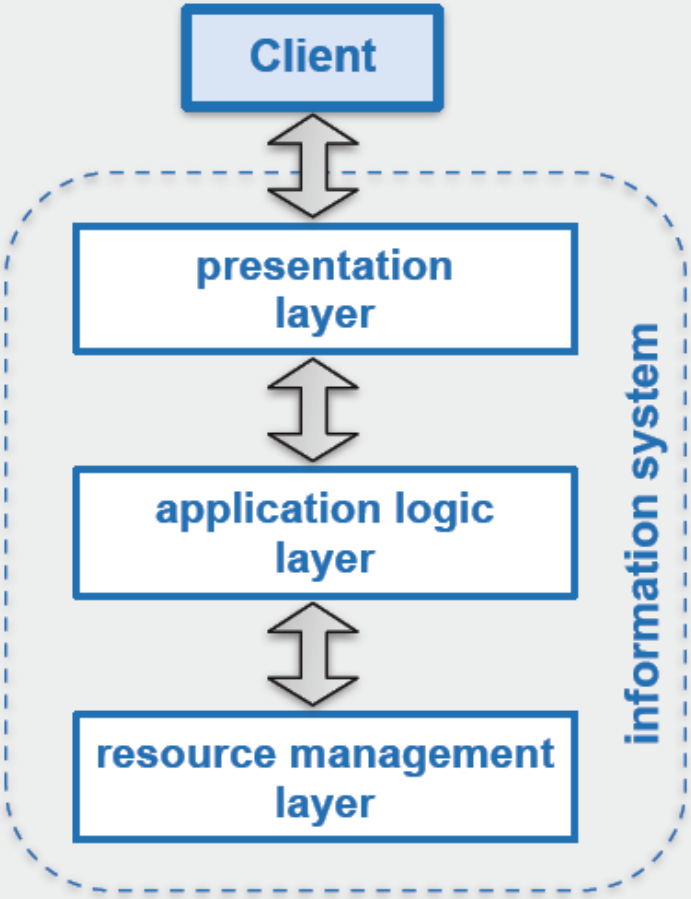
Resource Manager

- How ACID transactions are implemented
  - Allocate resources to a program executing a transaction (e.g., a locked record is a resource)
  - Reclaim resources in appropriate state on commit or abort
- Handled at the "Resource Management Layer"

Multi-Tier Architecture

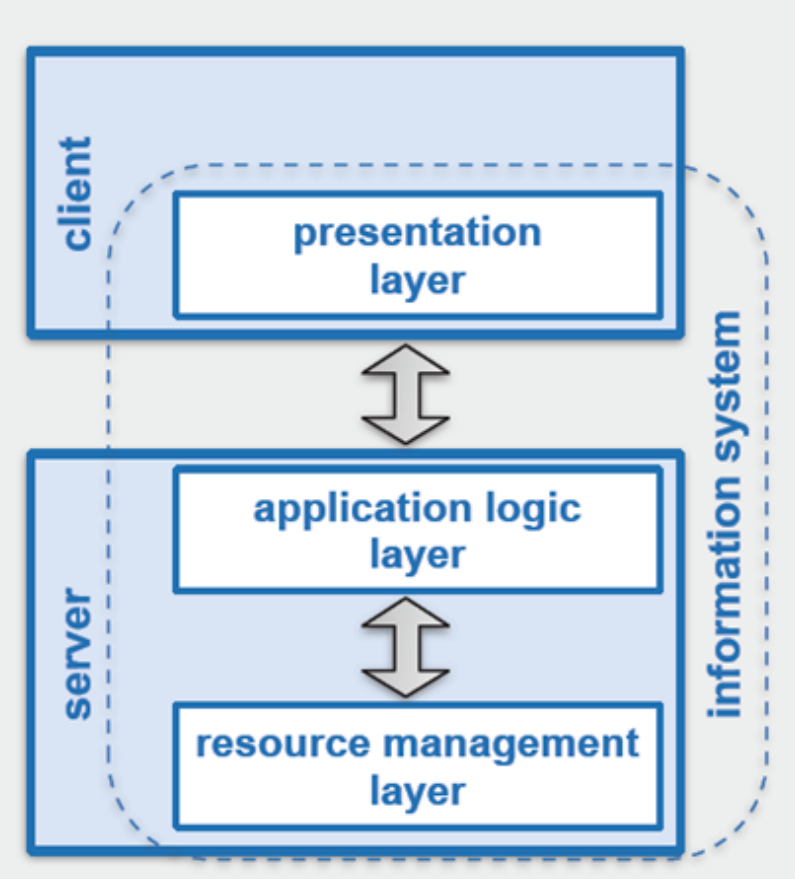
1-Tier Architecture

- System is necessarily monolithic
- May be highly efficient
- No stable service interface API
- Problem of Legacy Systems



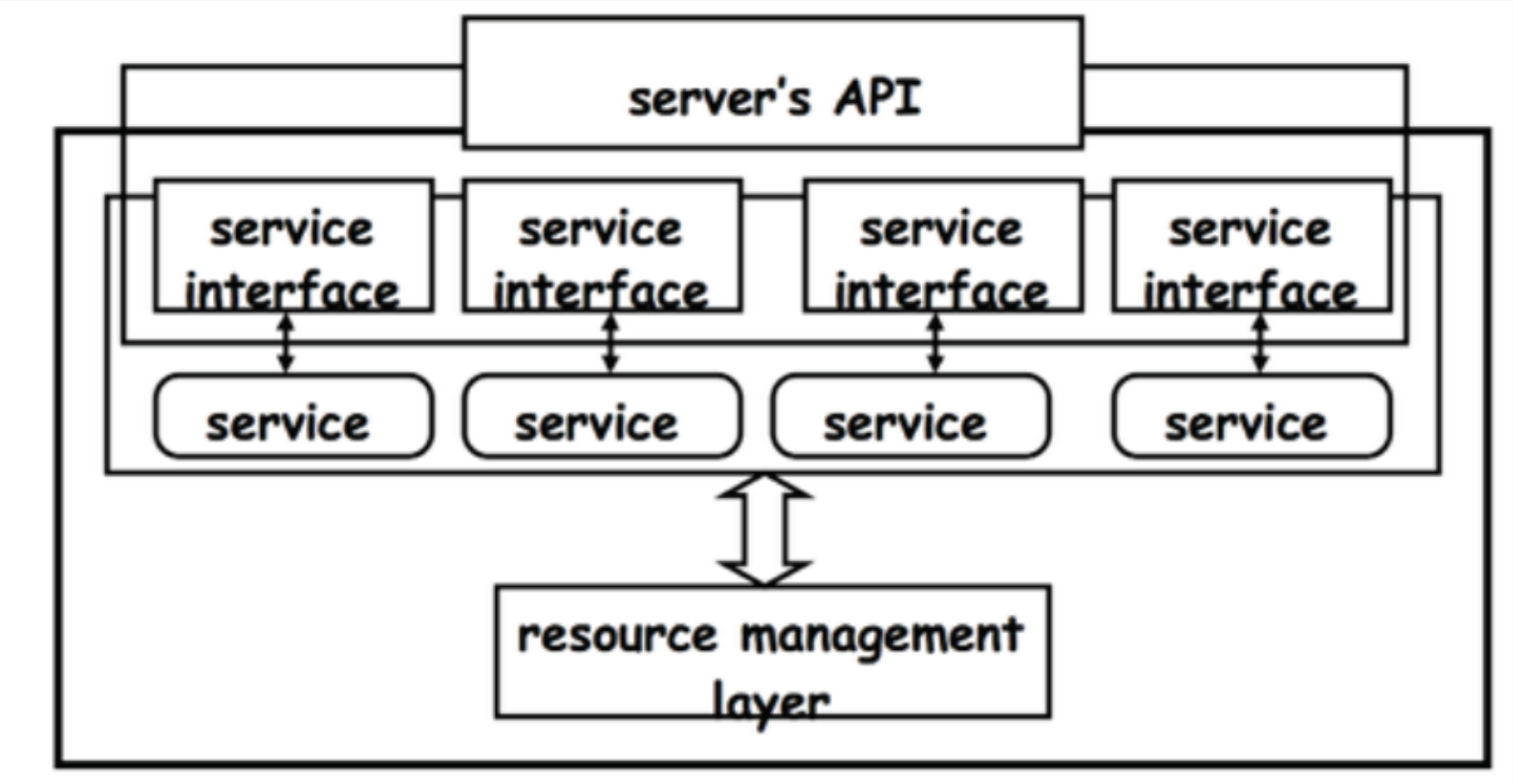
2-Tier Architecture

- Added flexibility in presentation layer
  - e.g., multiple specialized presentation layers add no complexity to application
- Encouraged stable, published APIs
  - so clients could be developed



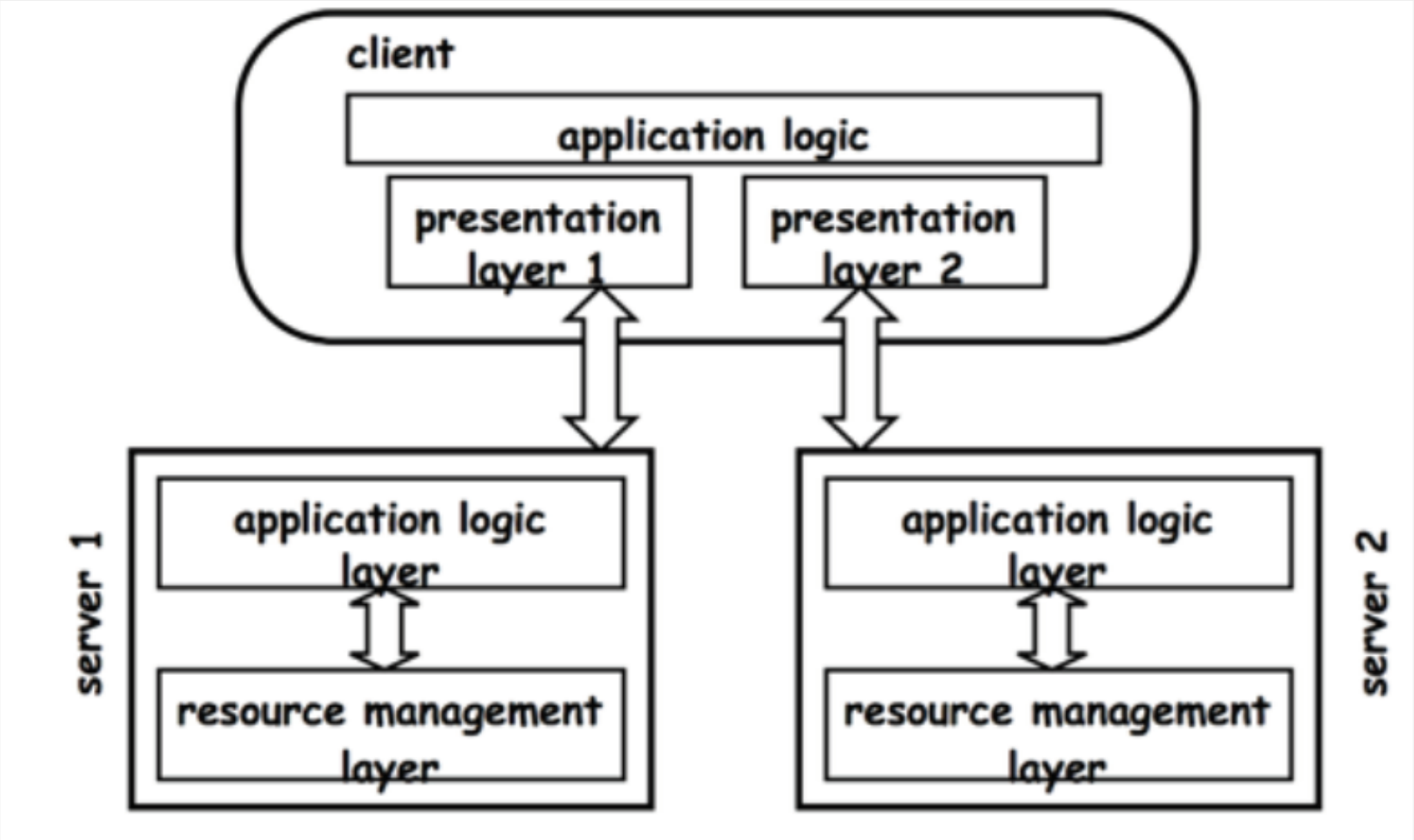
2-Tier Architecture Disadvantages

- A single server doesn't scale



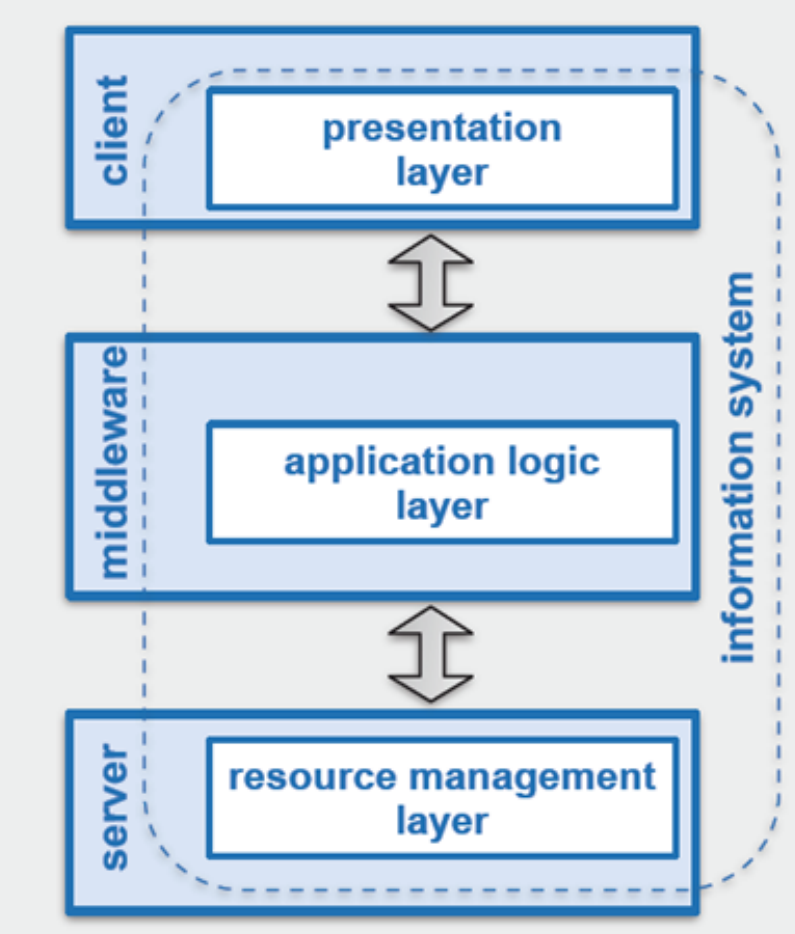
2-Tier Architecture Disadvantages

- Integration of multiple services must be done at client

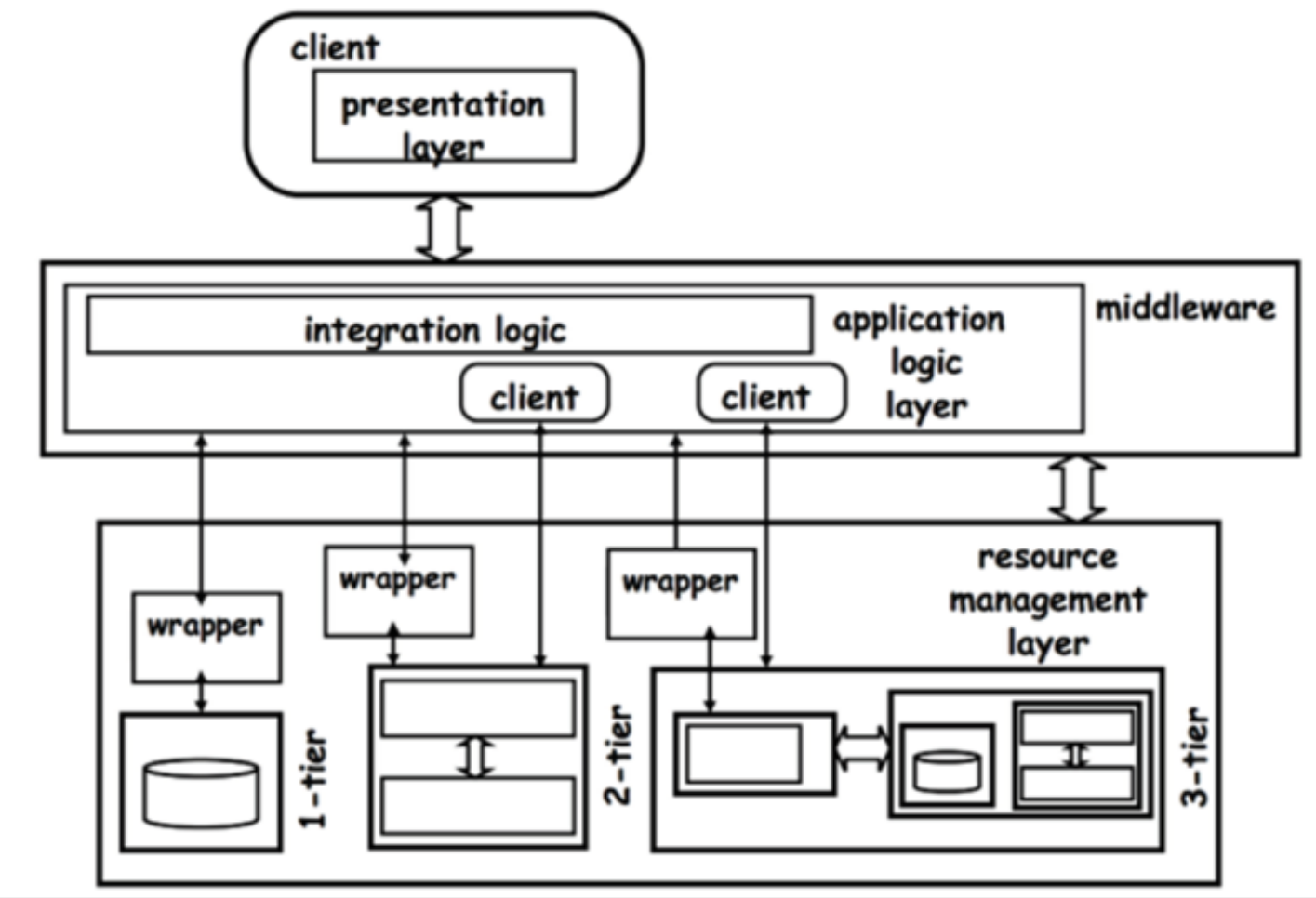


3-Tier Architecture

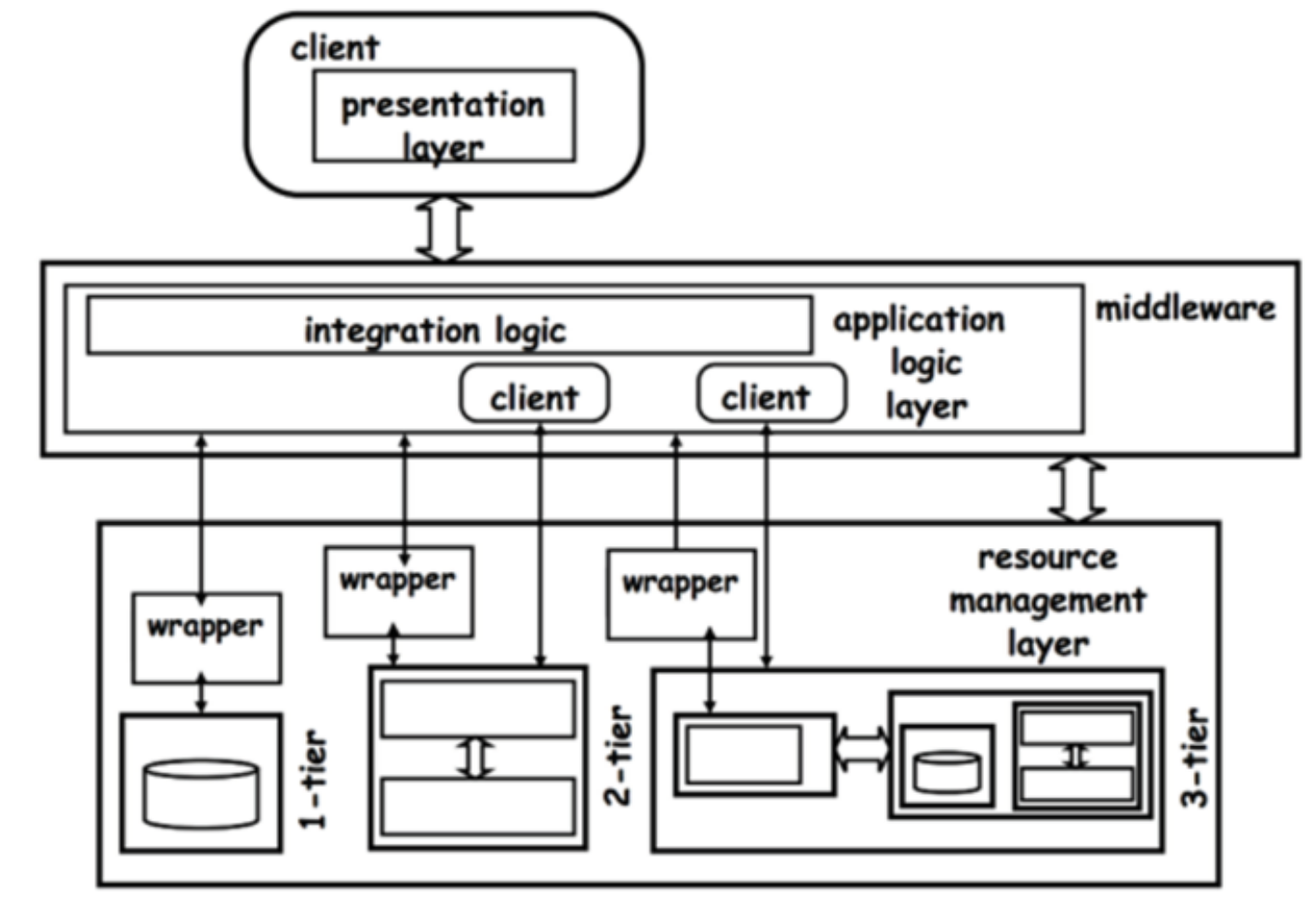
- Scalability at application Layer
  - Multiple application servers
- Application Integration
  - Do it in the middle tier
- Encourage stable, published APIs for resource management



Integration in the Middle Tier



N-Tier Architecture (Inductivity)



3-Tier 4-Tier 5-Tier

Enterprise Application Development Considerations

- Distributed computing protocols (RMI, CORBA, IIOP)
- Load balancing
- Persistence, back-end integration
- Transaction processing
- Clustering
- Runtime re-deployment, Server restarting
- Multi-threading
- Resource pooling
- Security, performance, optimization
- ...